

CONTACT INFORMATION	Computer Science Department Boston University MCS 105C, 111 Cummington Street Boston, MA 02215 E-mail: zczhu@bu.edu Website: https://cs-people.bu.edu/zczhu	
RESEARCH INTERESTS	Databases, Data Management and Data systems.	
EDUCATION	Ph.D. in Computer Science, Boston University, Massachusetts, USA.	2019 - Present
	MPhil in Computer Science, the University of Hong Kong (HKU). Thesis: “Effective and Efficient Discovery of Top-k Meta Path in Heterogeneous Information Networks”	2017 - 2019
	B.Sc. in Software Engineering, Tsinghua University.	2013 - 2017
PROFESSIONAL EXPERIENCE	• PhD Researcher at Boston University • Software Engineering Intern at RocksDB, Meta • Research Intern at DASLab SEAS, Harvard • System Engineer Intern at Weibo Corporation	Sept 2019 - Present May 2022 - Aug 2022 July 2018 - Oct 2018 June 2017 - Aug 2017
ACADEMIC SERVICE AND AWARD	• ACM SIGMOD Student Travel Award 2022 • ACM SIGMOD Availability Committee 2021 • Online student volunteer in VLDB 2020 • Committee member in ICTAI 2021 • Student volunteer in BDAI Workshop, HKU, 2019 • External reviewer of WISE2020, ER2019, CIKM2018, DASFAA2018, DASFAA2022, DASFAA2023	
PUBLICATIONS	1. Zichen Zhu. <i>SHaMBa: Reducing Bloom Filter Overhead in LSM Trees</i>, PhD Workshop in VLDB, 2023 2. Subhadeep Sarkar, Tarikul Islam Papon, Zichen Zhu, Dimitris Staratzis, Manos Athanassoulis. <i>Enabling Timely and Persistent Deletion in LSM-Engines</i>, ACM Transactions on Database Systems (TODS), 2023 3. Siqiang Luo*, Zichen Zhu*, Xiaokui Xiao, Yin Yang, Chunbo Li, Ben Kao. (* Equal Contribution). <i>Multi-Task Processing in Vertex-Centric Graph Systems: Evaluations and Insights</i>, EDBT, 2023 4. Ju Hyoungh Mun, Zichen Zhu, Aneesh Raman, Manos Athanassoulis. <i>LSM-Tree Under (Memory) Pressure</i>, Proceedings of the International Workshop on Accelerating Data Management Systems Using Modern Processor and Storage Architectures (ADMS), 2022 5. Manos Athanassoulis, Subhadeep Sarkar, Tarikul Islam Papon, Zichen Zhu, Dimitris Staratzis. <i>Building Deletion-Compliant Data Systems</i>, IEEE Data Engineering Bulletin (DEBull), 2022	

	- Subhadeep Sarkar, Dimitris Staratzis, Zichen Zhu, Manos Athanassoulis. <i>Constructing and Analyzing the LSM Compaction Design Space</i>, In Proceedings of the VLDB Endowment, Vol. 14(11), 2021. Zichen Zhu, Ju Hyoungh Mun, Aneesh Raman, Manos Athanassoulis. <i>Reducing Bloom Filter CPU Overhead in LSM-Trees on Modern Storage Devices</i>, In Proceedings of the International Workshop on Data Management on New Hardware (DaMoN), 2021. Zichen Zhu, Tsz Nam Chan, Reynold Cheng, Loc Do, Zhipeng Huang, Hoaci Zhang. <i>Effective and Efficient Discovery of Top-k Meta Paths in Heterogeneous Information Networks</i>, In IEEE Transactions on Knowledge and Data Engineering, doi: 10.1109/TKDE.2020.3037218. - Stratos Idreos, Niv Dayan, Wilson Qin, Mali Akmanalp, Sophie Hilgard, Andrew Ross, James Lennon, Varun Jain, Harshita Gupta, David Li, Zichen Zhu. <i>Design Continuums and the Path Toward Self-Designing Key-Value Stores that Know and Learn</i>, Conference on Innovative Data Systems Research (CIDR), 2019. Zichen Zhu, Reynold Cheng, Loc Do, Zhipeng Huang, Hoaci Zhang. <i>Evaluating Top-k Meta Path Queries on Large Heterogeneous Information Networks</i>, In Proceedings of the International Conference on Data Mining (ICDM), 2018.
DEMOS	Zichen Zhu, Subhadeep Sarkar, Manos Athanassoulis. <i>Acheron: Persisting Tombstones in LSM Engines</i>, In Proceedings of the ACM SIGMOD International Conference on Management of Data, Demonstration track, 2023 - Subhadeep Sarkar, Kaijie Chen, Zichen Zhu, Manos Athanassoulis. <i>Compactionary: A Dictionary for LSM compactions</i>, In Proceedings of the ACM SIGMOD International Conference on Management of Data, Demonstration track, 2022 Zichen Zhu, Siqiang Luo, Xiaokui Xiao, Yin Yang, Dingheng Mo, Yufei Han. <i>VC-Tune: Tuning and Exploring Distributed Vertex-Centric Graph Systems</i>, IEEE International Conference on Data Engineering (ICDE), Demonstration Track, 2022.
TEACHING EXPERIENCE (SELECTED)	Boston University (Teaching Fellow) 2019 - Present - CS 561 (Data Systems Architectures) Spring 2022, 2023 - CS 660 (Graduate Introduction to Database Systems) Fall 2020, Fall 2021 - CS 460 (Introduction to Database Systems) Spring 2021 - CS 237 (Probability in Computing) Spring 2020 - CS 105 (Introduction to Databases and Data Mining) Fall 2019 University of Hong Kong 2017 - 2019 - CCST9047 (The Age of Big Data) Spring 2018 - CCST9047 (The Age of Big Data) Spring 2019
MPHIL THESIS	Zichen Zhu. <i>Effective and Efficient Discovery of Top-k Meta Path in Heterogeneous Information Networks</i>, MPhil Thesis, 2019, Department of Computer Science, University of Hong Kong. (under the supervision of Dr. Reynold Cheng) I worked on effective and efficient discovery of top-k meta paths under the supervision of Dr. Reynold Cheng, Professor, Department of Computer Science, the University of Hong Kong. In this work, I observe that the shortest meta path between two nodes cannot identify the most representative relation and I propose a ranking function that seamlessly integrates existing ranking functions into a multi-step searching framework and I combine bidirectional searching with this framework to further boost the searching efficiency.

TECHNICAL
SKILLS

- **Programming Languages:** C, C++, Java, Python, JavaScript
- **Markup Languages:** HTML, XML, $\text{\LaTeX}$
- **Database Management Systems:** RocksDB, MongoDB, MySQL, PostgreSQL
- **Web Development:** Django, Flask, CodeIgniter

LAST UPDATE: June 2023